

Evaluation of pathological alpha-synuclein in skin biopsies as potential biomarker for Parkinson's disease

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Summary

Biomarkers to detect and quantify pathological alpha-synuclein (aSyn) in accessible samples from Parkinson's disease (PD) patients are urgently needed. We have developed aSyn seed amplification assay (SAA) for skin biopsies. Here, we measured 96 skin biopsy homogenates from prodromal, early and late PD individuals from the PPMI cohort. Homogenization of skin biopsies was performed by Indiana University.

Method

aSyn SAA

Human recombinant full-length (1-140 aa) alpha-synuclein without tag was expressed in *E. coli* and purified using standard purification methods. Alpha-synuclein from one single purification batch was used throughout this study. Directly before use in SAA, alpha-synuclein was centrifuged for 30 min at 21,000 xg (4°C). A mastermix containing 44.4 mM sodium phosphate (pH 8.2), 0.0167 % Triton X-100, 11.1 µM Thioflavin T (ThT), and 0.11 mg/ml alpha-synuclein was prepared. To each well of a black 384-well plate (Greiner bio-one, Germany), 45 µl of mastermix and 6.5 ± 1 mg of 0.1 mm zirconia/silica beads (Roth, Karlsruhe, Germany) were added. Skin samples were tested in two different dilutions with 4 technical replicates each. Five µl of pre-diluted skin lysate (pre-dilution in 100 mM sodium phosphate (pH 8.2)) were added per well yielding a final reaction volume of 50 µl/well and final skin lysate dilutions of 1:20 and 1:50. The final reaction mixture (50 µl/well) contained ~50 mM sodium phosphate (pH 8.2), 0.015 % Triton X-100, 10 µM ThT, and 0.1 mg/ml alpha-synuclein. The plates were sealed with a plate sealer film (Fisher Scientific Ltd, UK) and incubated in a BMG FluoSTAR Omega plate reader at 32 °C for 70 h with shake/rest cycles: 1 min double orbital shaking at 200 rpm, 14 min rest. ThT fluorescence measurements (450 nm excitation and 480 nm emission; top read) were taken every 15 min.

Before any further analysis, each aggregation curve was baseline corrected by subtracting the mean of the relative fluorescence units (rfu) measured between 2 and 5 h from all data points of the curve. A threshold of 3500 rfu was used to determine the time to threshold (TTT), i.e. the time when the curve reaches this threshold. If the threshold was not reached during the 70 h runtime of the assay, the TTT was set to 70 h. Curves that reached the threshold were defined as positive replicates. A sample was defined as SAA positive if 3 or 4 replicates reached the threshold within 70 h. If a sample reached the threshold in 2 out of 4 replicates, then the run was

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repeated for this sample. For such samples, the results of the second assay run were reported in the results file. If the second assay run showed again 2 out of 4 positive replicates, then the sample was rated as positive. In addition, the maximum fluorescence (Fmax) was determined as the highest rfu value recorded during the 70 h runtime of the assay. The area under curve (AUC) was calculated between 0 and 70 h. Negative AUC values that can occur due to minimal baseline drift were set to 0.

Table 1: Overview of the terms used in the results file.

TESTNAME	Description
SAA_1:20_status SAA_1:50_status	Final result indicating if a skin sample was SAA positive (1) or negative (0) based on 4 technical replicates.
SAA_1:20_TTT (rep 1-4) SAA_1:50_TTT (rep 1-4)	Time to threshold (TTT) in h for every sample and for each of the 4 technical replicates.
SAA_1:20_Fmax (rep 1-4) SAA_1:50_Fmax (rep 1-4)	Maximum fluorescence (Fmax) in rfu for every sample and for each of the 4 technical replicates.
SAA_1:20_AUC (rep 1-4) SAA_1:50_AUC (rep 1-4)	Area under the curve (AUC) for every sample and for each of the 4 technical replicates.

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